

CBSE Test Paper 01
Chapter 5 Laws of Motion

1. A monkey of mass 40 kg climbs on a rope which can stand a maximum tension of 600 N. What is the tension in the rope if the monkey falls down the rope nearly freely under gravity? **1**
 - a. 400 N
 - b. 200 N
 - c. 0 N
 - d. 600 N

 2. A constant retarding force of 50 N is applied to a body of mass 20 kg moving initially with a speed of 15 m s^{-1} . How long does the body take to stop? **1**
 - a. 7.0 s
 - b. 4.0 s
 - c. 6.0 s
 - d. 5.0 s

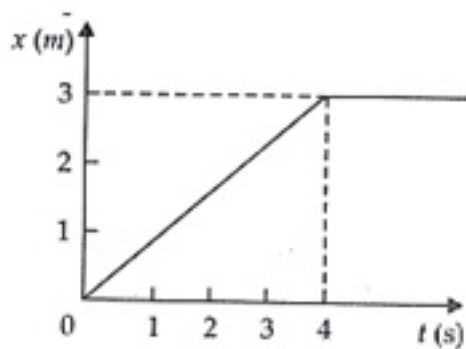
 3. Rolling friction is **1**
 - a. greater than static friction
 - b. less than static friction
 - c. greater than or equal to static
 - d. equal to static friction

 4. A nucleus is at rest in the laboratory frame of reference. If it disintegrates into two smaller nuclei **1**
 - a. the products must move at an angle in same direction.
 - b. the products must move in same direction.
 - c. the products must move at an angle in opposite directions.
 - d. the products must move in opposite directions.

 5. force is required **1**
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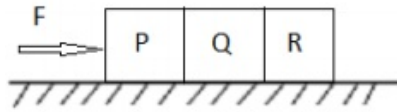
- a. only to keep an object moving
- b. only to stop a moving object
- c. to start a stationary object and to stop a moving object
- d. only to start a stationary object moving

6. Does inertia of a body depend on any factor other than mass? **1**
7. Why is it desired to hold a gun tight to one's shoulder when it is being fired? **1**
8. If force is acting on a moving body perpendicular to the direction of motion, then what will be its effect on the speed and direction of the body? **1**
9. The position-time graph of a body of mass 2 kg is as given in Figure. What is the impulse on the body at $t = 0$ s and $t = 4$ s? **2**



10. How does banking of roads reduce wear and tear of the tyres? **2**
11. What do you mean by inertia of motion? Give an example to illustrate it. **2**
12. In a circus a motorcyclist driving in vertical loops inside a 'death-well' (a hollow spherical chamber with holes, so the spectators can watch from outside). Explain clearly why the motorcyclist does not drop down when he is at the uppermost point, with no support from below. What is the minimum speed required at the uppermost position to perform a vertical loop if the radius of the chamber is 25 m? **3**
13. A 70 kg man stands in contact against the inner wall of a hollow cylindrical drum of radius 3 m rotating about its vertical axis with 200 rev/min. The coefficient of friction between the wall and his clothing is 0.15. What is the minimum rotational speed of the cylinder to enable the man to remain stuck to the wall (without falling) when the floor is suddenly removed? **3**

14. Three identical blocks each having a mass m , are pushed by a force F on a frictionless table as shown in figure



What is the acceleration of the blocks? What is the net force on the block P? What force does P apply on Q. What force does Q apply on R? 3

15. A body of mass 400 gm moving initially with a constant speed of 36 km h^{-1} towards the north is subjected to a constant force of 8.0 N directed towards the south for half of a minute. Beyond that time the body continues its motion with uniform velocity. No other forces are acting on the body throughout its motion. Take the instant the force is applied to be at $t = 0 \text{ s}$, the position of the body at that time to be $x = 0 \text{ m}$. Find out its position at $t = -5 \text{ s}$, 25 s and 100 s respectively applying Newton's equations of motion.